

Skills Summary

Finding and Verifying Inverse Functions

Use this reference while completing your worksheet or when studying.

What an inverse is. f^{-1} undoes f : if $f(a) = b$ then $f^{-1}(b) = a$. So $f^{-1}(f(x)) = x$ and $f(f^{-1}(x)) = x$.

Notation warning. f^{-1} is the inverse *function*. It is not $\frac{1}{f(x)}$.

Domain and range swap. The domain of f^{-1} is the range of f , and its range is the domain of f . Graphically, f^{-1} is the reflection of f across the line $y = x$.

Only one-to-one functions have inverses. If a horizontal line meets the graph of f twice, two inputs share an output and there is nothing for f^{-1} to return — restrict the domain first.

1. Find the inverse: swap, then solve

Write $y = f(x)$; swap x and y ; solve the new equation for y ; rename it $f^{-1}(x)$. Solving means undoing the operations **in reverse order** — whatever was done last to x is undone first.

Example: Find the inverse of $f(x) = 4x + 3$.

Step 1. The question asks which input produces a given output, so swap the roles of the variables and solve for the new y .

Step 2. $y = 4x + 3$ becomes $x = 4y + 3$. Subtract 3: $x - 3 = 4y$. Divide by 4: $y = \frac{x - 3}{4}$.

$$f^{-1}(x) = \frac{x - 3}{4}$$

Try it: Find the inverse of $f(x) = 6x - 1$.

$$\frac{9}{1+x} = (x)_{1-f} \text{ check}$$

2. Verify a pair with composition

Two functions are inverses only when **both** $f(g(x)) = x$ and $g(f(x)) = x$. Substitute one whole function into the other and simplify; the variable must survive alone, with every constant cancelling.

Example: Are $f(x) = \frac{x}{2} + 4$ and $g(x) = 2x - 8$ inverses?

Step 1. “Verify” means composition, not re-deriving — substitute g into f and simplify, then repeat the other way round.

Step 2. $f(g(x)) = \frac{2x - 8}{2} + 4 = (x - 4) + 4$, and the constants cancel. The composition collapses to x , and $g(f(x))$ does the same, so the pair is inverse.

Try it: Simplify $f(g(x))$ for $f(x) = 3x - 1$ and $g(x) = \frac{x+1}{3}$. What does it come out to?
 check: x — so they are inverses

3. Inverse values, domain and restriction

$f^{-1}(b)$ asks: which input of f gives the output b ? You can answer it either by building the formula for f^{-1} and evaluating, or by solving $f(x) = b$ directly — the two routes must agree. When f is not one-to-one, restrict its domain first (for $f(x) = x^2$ take $x \geq 0$); the inverse then inherits its domain from the range of the restricted f .

Example: For $f(x) = 2x + 7$, find $f^{-1}(9)$.

Step 1. Only one value is wanted, so either build the inverse formula or solve $f(x) = 9$ — the formula route also leaves you something reusable.

Step 2. $x = 2y + 7$ gives $f^{-1}(x) = \frac{x-7}{2}$, so $f^{-1}(9) = \frac{9-7}{2} = \frac{2}{2}$. $f^{-1}(9) = 1$, and indeed $f(1) = 2 + 7 = 9$.

Try it: For $f(x) = 5x - 2$, find $f^{-1}(13)$. Check your answer by putting it back into f .
 check: $f(3) = 13$

(!) Watch out: undo operations in reverse order. For $f(x) = 3x + 5$ the inverse is $\frac{x-5}{3}$, not $\frac{x}{3} - 5$ — the 5 comes off before the division, because it was added last.